AKADEMIKA

FOL-CSK

CONNECTOR AND SPLICE INSTALLATION KIT

EXPERIMENTAL MANUAL

…. A MESSAGE FROM

AKADEMIKA

Today's system designers are faced with tomorrow's problems. FIBER OPTIC COMMUNICATION is one of the important subject need to teach while learning electronics.
It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO;

- Light years ahead – refers to leadership.

As leaders in our industry in India, We are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that We have a complete, truly innovative & superior training products we are so committed to quality that we back our products with a complete comprehensive warranty.

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SAFETY RULES

Carefully follow the instructions contained in this handbook, as they provide you with important points on safety during the installation, use and maintenance. Keep this handbook at hand for any further help.

Set all accessories in order on unpacking, and check integrity with respect to its checklist. Check the Kit for any visible damage.

Before connecting the power supply to the equipment, be sure patch cords are connected correctly, jumpers are correct as per experimental position.

In **FOL-CSK KIT**, signals are to be monitored with an oscilloscope, the scope input channel should be a.c. coupled, unless otherwise indicated. Ensure that X10 oscilloscope probes are used throughout.

These kits must be employed only for the use for which it has been conceived, i.e. as educational kits, and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of fault or malfunctioning of the trainers, turn off the power supply and do not tamper the kits. In case servicing is required, contact the service center for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

These kits are warranted against defects in workmanship and materials. If any failure resulting from a defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting patch chords, Crystals, etc. and other imported items.

This warranty is subject to the following conditions and limitations. The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on Akademika Lab Solutions sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the event warranty service is needed, purchaser should get in touch with service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kit /system has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika Lab Solutions judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs and maintenance of old and decrepit products, be sure to contact our service center or sales outlet.

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FUNCTIONAL BLOCK

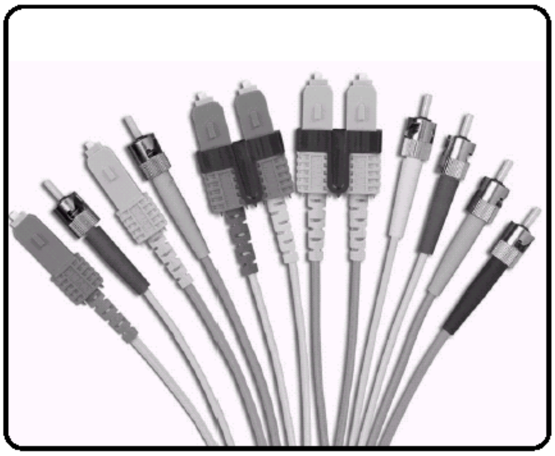
INTRODUCTION

In any fiber optic system, one will always find the need for splices and connectors. Splices are required during installations and again during service, if the cables were damaged and fibers become broken. Connectors are invariably used to connect fiber to the terminal equipment. Each splice or connector give rise to additional attenuation and the need to minimize such losses requires a careful preparation of the ends of the fibers. Low loss is obtained only if the fiber end faces are clean, smooth and perpendicular to the fiber axis.

Connectorisation

It is the process of terminating the fiber by a connector. Normally a source and detectors are mounted is a standard bulkhead, into which the fiber link of the system, which is having the connector at the end, is plugged such an arrangement permits individual units and components to be tested separately and replaced if necessary. Moreover this arrangement also allows maximum coupling of light.

Many different designs of connectors have been developed. (fig.I) Most use a ferrule for fiber alignment, although some designs use three spheres to center the fiber and a double cone to align the connectors. Some of the more sophisticated types, intended for use in high-speed systems where back-reflections must be minimized, allow physical contact between the two fibers.



The SMA connector is commonly used with multimode fibers. It has a 9mm long ferrule with a precision bore designed to match the cladding diameter and a standard SMA-threaded sleeve to make the connection. A typical insertion loss is in the region of 1.0dB initially, but mechanical wear and the likelihood of the fiber end surfaces becoming contaminated and abraded in working environments during life which may involve hundreds or thousands of disconnections and re-connections causes this loss to increase.

The ST connector is intended for use with multimode or mono-mode fiber. It is relatively easy to terminate and employ a bayonet-coupling collar somewhat similar to the BNC co-axial connector. Typical insertion losses of 0.5dB can be obtained in clean conditions.

Some relatively inexpensive, bi-conical, plastic, snap-in connectors have beer developed, of which the type SC is one. Another standard is the two-fiber connector specified for the FDDI high capacity local area network.
The FC connector is an example of one that is suitable for use with mono-mode or multimode fibers in high capacity systems. Insertion losses are normally less than 0.5dB. It is available in PC (Physical contact) and APC (angled physical contact) forms for those applications that may be very sensitive to low levels of back-reflection. These versions also give lower coupling losses, typically 0.2dB. It is normal to use a precision-machined ferrule with a ceramic capillary to act as the fiber guide, and an antirotation mechanism is incorporated.
In connectorisation process, first of all the glass fiber end is prepared so that the fibers end faces are clean, smooth and perpendicular to fiber axis. Then the glass fiber is secure in the connector housing using epoxy adhesive and made permanent by crimping the housing.

Splicing
It is the process of joining two pieces of fiber together. On a typical fiber optic link there are usually a number of splice points as it is typically not possible (or practical) to simply “pull” one long continuous fiber. Splice quality is absolutely critical, because each junction point on the link introduces some degree of optical power loss or attenuation. Splicing joints the two fibers together both optically and physically.

Splicing requires a high degree of operator training. Extreme care must be taken to ensure that the fiber is spliced together in a clean "dirt free" environment. Even a small amount of contamination can lead to poor splice quality. The ends of the fiber must be flat and smooth with no cracks, chips, gouges or other physical defects. Specialized crimping/cleaving tools must be used to ensure accuracy. The cleaving tool is used to score/scribe the glass, which is then broken off to obtain as good a surface profile as possible. Cleaving is perhaps the most important part of the splicing process. If the ends of the fibers are not uniform, the splice results will be poor. A field microscope is often used to inspect the quality of the surface. Once the fiber has been properly stripped and cleaved, special alcohol pads are used to clean the surface in preparation for the final splicing.

Apart from the cleaving process, other factors can influence splice quality such as:
(i) **Core Mismatch** -the core diameters of the two fibers are different
(ii) **Lateral Core Offset** -the core is not centered within the fiber.
(iii) **Angular Misalignment** -the ends of the fibers must be less than 2 degrees apart.
(iv) **Excess End Separation** -the two fiber ends are positioned too far apart.
(v) **Contamination By Dirt Or Dust**.

There are essentially two different techniques that are currently in use, fusion and mechanical splicing. Fusion splicing involves the fusing of the two fiber ends together using high temperature, which is typically achieved through use of high voltage electrodes. The two ends of the fiber must be aligned very precisely before they are fused together. Most fusion splicing equipment either does this automatically or provides controls to make very small adjustments. The actual fusing of the ends only takes a few seconds. After the fiber ends have been fused together, a protective splice protector is attached to protect the splice and provide stain relief.

Fusion splicing is the most effective splicing technique, and when done properly should result in less than 0.5dB of loss across the splice Even greater accuracy can be obtained, if very expensive “lab type" fusion equipment is used. Fusion splicing equipment is typically very expensive and not very portable. In field environments this type of equipment is typically used inside a truck.
Mechanical splicing is not quite as accurate as fusion splicing, as the two ends are not physically “fused" together. In this technique, the protective splice cover is used during the splicing process. Some kind of curing agent, typically U/V fluid or some kind of heat sensitive fluid is injected into the splice cover. The fluid is then “cured" inside the splice cover thus locking the two fiber ends in place. Mechanical splices can also be used to make temporary repairs if index-matching fluid is used instead of a curing compound. This is particularly useful for quick repairs of broken cables. Mechanical splicing does not require very expensive fusing splicing equipment. It is a low cost alternative in field application typical losses across the splice are less than one dB.

FOL-CSK- St Connector and Splice Installation Kit

FOL-CSK- Connector & Splice installation kit contains all the tools and material required to terminate glass fiber with ST Connector and joint glass fiber with Ultra Splice.
The kit consists of following items:
ST Polishing Tool
Fiber Polishing Sheets
Polishing Pad
Fiber Optic Microscope
Diamond Scribe
Jacket Stripper
Buffer Stripper
Crimp Tool
Blade and Scissors
Optic Prep
Cotton Swabs
Disposable Syringe with Needle
ST Connector
Epoxy pack
Ultra Splice
The purpose of this kit is to train the students in the technique of connectorisation. Before starting with the procedure, let us discuss in brief the operation and application of the above tools.

ST Polishing Tool
It helps to hold the ST Connectorised fiber at 90^0 while polishing on the Fiber Polishing sheet.

Fits most 2.5 mm ferrule connectors including: ST, SC and FC connectors
Inside and outside rims designed for improved hand positioning while performing the polishing operation
Channeled, raised base allows debris to escape easily and provide for easy polishing action

Fiber Polishing Sheet
The polishing sheet comes with fine grain polishing paper. For polishing the fiber, rotate the ST polishing tool in the figure of eight on polishing paper.

Polishing Pad
The Polishing pad is used to hold polishing paper. It provided proper surface for connectorisation.

Fiber Optic Microscope
It is used to observe the tip of fiber.

Zoom Pocket Microscope is perfect for fiber optic end termination inspection its End design set up to accept 2.5 mm ferrule connectors with Easy to adjust focusing wheel

1. Unfold the battery case so that the built in lamp will light up.
2. To observe the tip of the bare fiber, set the fiber tip to the hood center. If the fiber is ST connectorised replace the hood with the ST Adapter and hold the ST Connector in the adapter
3. Leek into ocular lens and adjust focus, turning the focus adjust knob.

Diamond Scribe
It is use to cut the glass fiber. Hold the diamond scribe perpendicularly to the fiber axis and slowly scribe it on the glass fiber without exerting any pressure.

Fiber Optic Stripper
It is designed to remove the outside jacket of the fiber. To use it insert the fiber into the stripping hole of proper diameter. Close the tool and draw fiber through hole exerting steady pressure.(refer fig.)

For stripping 250 micron buffer coating from 125 micron optical fiber its Precision diameter hole & V-opening in blade allow for accurate buffer coating removal

Crimp Tool
After the fiber has been bonded permanently to the connector, the back shell (crimp sleeve) of the connector is crimped using crimp tool. Select proper cavity diameter of crimp tool while crimping the connector.

Blade and Scissors
They are used to remove the outer jacket and to cut the fibrous (kevlar) material of the fiber.

Optic PREP & Cotton Swabs
They are used for cleansing purpose. For minimizing the loss, one should ensure that no dust particles are adhered to the bare fiber Wipe the fiber with Optic prep to remove the impurities.

Epoxy
It is used for bonding the fiber inside the connector permanently. It consists of two parts. Take equal amount of adhesive from each part on any clean, dry, flat surface. Mix it thoroughly so that the different colored part blend into a smooth uniform color. After applying allow it to cure for 4 hours. Cure may be accelerated by heat such as one hour at 140^0F. Note that the mixed material must be used within 30 minutes.

Disposable Syringe with Needle
Syringe is used to insert Epoxy into connector ends.

EXPERIMENT NO.1

EXPERIMENT NO-1

NAME

Connectorisation of Jacketed Glass Fiber

AIM

To terminate one meter of jacketed glass fiber with ST Connector at both the ends

EQUIPMENTS

1 meter Jacketed Glass Fiber
FOL-CSK Connector and Splice installation Kit

PRECAUTIONS

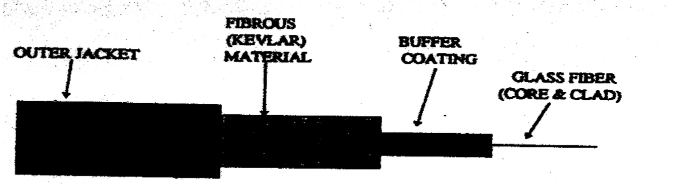
Please note the following instructions carefully

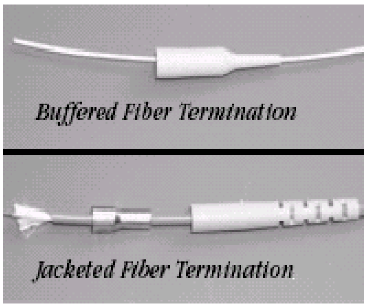
- The Connectorisation Kit contains precision mechanical items. Handle the tools with great care.
- Avoid direct skin contact with the chemicals. Irritation and sensitization might occur on contact, wash areas immediately with water. Do not use solvents for washing in case of eye contact. Wash with water for at least Fifteen minutes and get medical attention.
- Carry out the experiment in a dust-free & dry environment. During the process, absolute care should be taken to keep the bare fiber free from impurities, in order to minimize losses.
- Be careful to dispose of the chips of glass fiber while cutting.

PROCEDURE

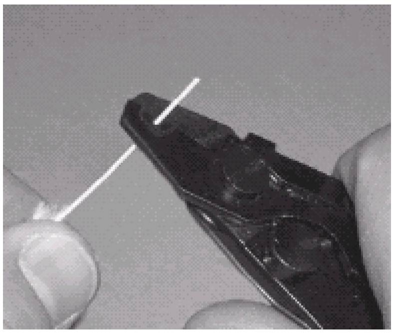
- Cut one meter of jacketed glass fiber optic cable. The fiber consists of core & clad of glass fiber, buffer coating, fibrous (Kevlar) material and outer jacket.
- Place the conical strain relief sleeve over the end of fiber such that broader end of sleeve should be kept towards the fiber tip.

STRUCTURE OF A TYPICAL JACKETED GLASS FIBER





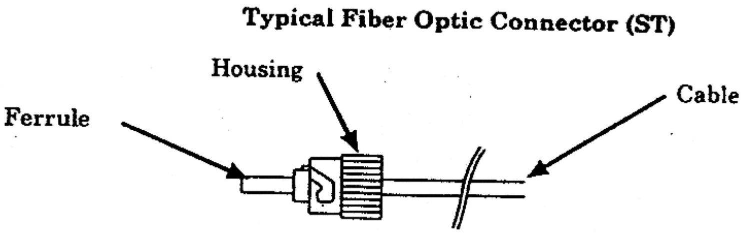
- Remove the outer jacket coating of about ½ inch using fiber optic jacket stripper.
- Cut the fibrous (Kevlar) material using blade or scissors.
- Remove ½ inch of buffer coating using the fiber optic buffer stripper. Select the stripper hole of smaller diameter while removing the buffer. The buffer coating can also be removed chemically. Dip the fiber end in Methylene Chloride so that the coating get dissolved.



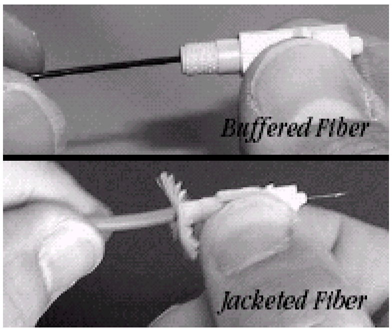
CAUTION: Care should be taken to avoid cutting the bare fiber of while trying to remove the outer and buffer coating.

- Wipe the bare fiber end with optic prep to remove the dust particles, which may have adhered to the fiber. To minimize losses one should ensure to remove all traces of impurities.

- Insert the fiber end into the connector and ensure that bare fiber emerges out from the ferrule of the connector smoothly. Remove the fiber from the connector.
- Take an equal amount of resin from each part of the epoxy and mix it thoroughly. Apply a small amount of epoxy inside the housing of the connector and fill it completely using syringe with needle.



- Insert the stripped fiber into the connector and gently manipulate the cable without exerting any pressure so that the bare fiber finally emerges from the ferrule. Ensure that all the extra part of bare fiber should protrude out from the ferrule.



- Clean the connector with cotton swabs to remove the additional epoxy that has come out of the connector.
- Allow the epoxy to cure for four hours. Take care that the position of the fiber remains unchanged during this process. The fiber is not fixed in that position inside the connector. Curing process can be accelerated by heating at 140^0F for about one hour. One can use Hair dryer for this purpose.

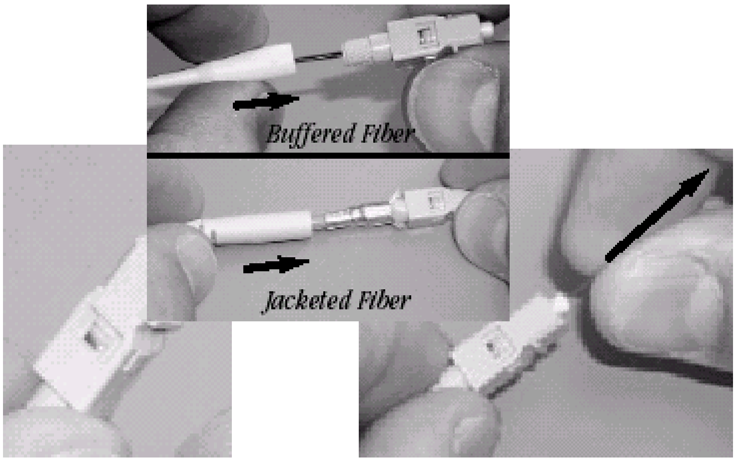
CAUTION

i. Care should be taken while applying epoxy to prevent any of it from reaching the connector ferrule. This would block the hole in the ferrule face and render the connector useless.

ii. Do not apply pressure while inserting the fiber in the connector. If the fiber breaks, pieces of glass could clog the hole in the connector ferrule.

- Crimp the back shell of the connector using the crimp tool. Care should be taken for selecting the proper cavity diameter.

- Slide the conical protective sleeve over the housing of the connector.



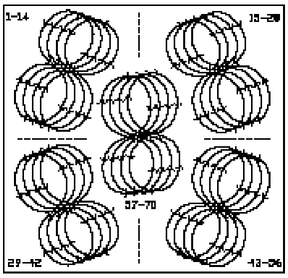
- Cut the bare fiber part emerging out of the ferrule using the diamond scribe. Extreme care should be taken while scribing and ensure that the cut is made perpendicularly and about 1 mm of the fiber still protrude out of the ferrule.

- Wipe the tip of the ferrule with optic prep.

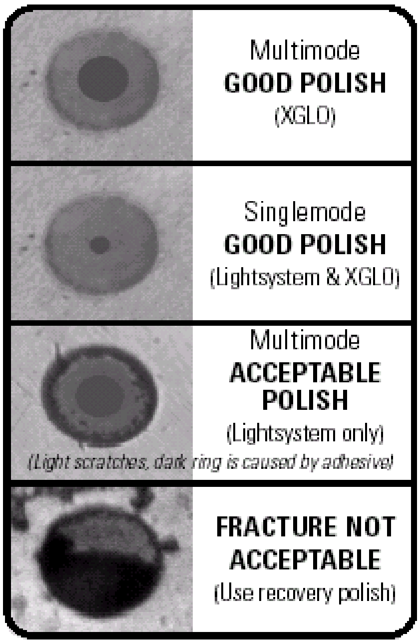
- Clean the Polishing pad and Polishing tool using tissue paper.
- Place one sheet of polishing paper on the pad, taking care to avoid wrinkles and bubbles on the sheet. If these sheets are not being used for the first time, ensure that there are no remnants of glass or other impurities on them. Clean them with tissue paper if necessary.

- Hold the connectorised fiber inside the ST polishing disc. This will ensure that the fiber is held at 90^0 while polishing. Rotate the disc over the polishing sheet in the manner of figure of eight pattern. For about 40-50 times or more than that if required.



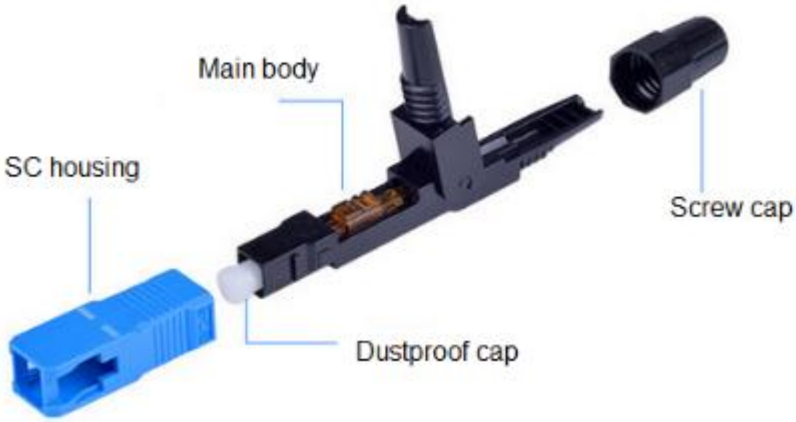


- Observe the tip of the fiber under Microscope. Switch on the lights of microscope. Concentrate the light beam on the ferrule. Turn the focusing wheel & zoom switch for better visibility. One can view two concentric circle (core and clad) along with some dark spot.
- Polish the fiber until there are no dark spot observed under the microscope. The core and the clad regions should be clearly visible. The absence of this would mean that either the fiber has slipped inside the ferrule while polishing as the bonding was inadequate or the fiber end has broken inside ferrule, in which case the complete process should be repeated. The perfect mirror finish will ensure maximum coupling of light through the fiber.

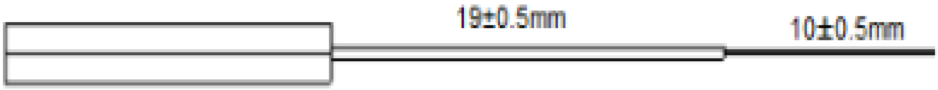


- Repeat the above procedure to terminate the other end of the fiber. The ST connectorised patch chord can now be used as a link between an optical transmitter and a optical receiver. Another patch chord can be connected to this patch chord using an adapter.

SC Type Fast Connector Termination Process.



Step 1: Lead the cable into the screw cap.
Step 2: Strip the cable around 50mm with the drop cable tripper; Peel off the fiber coating with a Miller pliers. Clean the fiber with alcohol and dust-free paper, cut the fiber as below length.



Step 3: Insert and press the cable into the groove and keep the fiber slightly bending. Push the fixed ring forward, close the gland for fixing the drop cable.

Step 4: Close the screw cap, put SC housing and finish assembling.

EXPERIMENT NO.2

EXPERIMENT NO-2

NAME

Measurement of Connectorisation Loss

OBJECTIVE

To Study of Measurement of Connectorisation Loss

EQUIPMENT

- 1 meter Standard Glass Patch chord
- Optical Power Source
- Optical Power Meter

PROCEDURE

- Take one meter standard glass patch chord
- Connect one end of the patch chord to optical power source and other end to optical power meter.
- Switch on the power source and power meter. Select the wavelength in power meter in accordance with the wavelength of the source.
- Note the reading in power meter.
- Replace the standard patch chord with the patch chord under test.
- Note the reading in power meter and find out the difference of the readings.
- The difference will denote connector loss.

EXPERIMENT NO.3

EXPERIMENT NO-3

NAME

Splicing of Jacketed Glass Fiber

AIM

To join two jacketed glass fiber cables using Ultra Splice.

EQUIPMENTS

- Two pieces of Jacketed Glass Fiber
- FOL-CSK Connector
- Splice installation Kit

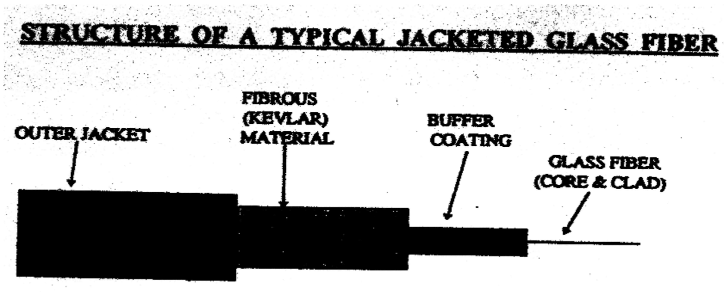
PRECAUTIONS

Please note the following instructions carefully

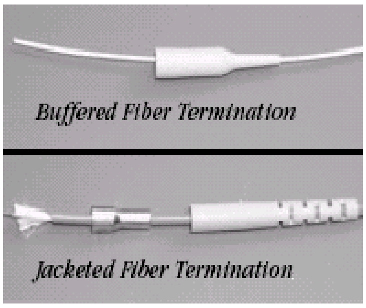
- The Splicing Kit contains precision mechanical items. Handle the tools with great care.
- Avoid direct skin contact with the chemicals. Irritation and sensitization might occur on contact, wash areas immediately with water. Do not use solvents for washing in case of eye contact. Wash with water for at least Fifteen minutes and get medical attention.
- Carry out the experiment in a dust-free & dry environment. During the process, absolute care should be taken to keep the bare fiber free from impurities, in order to minimize losses.
- Be careful to dispose of the chips of glass fiber while cutting.

PROCEDURE

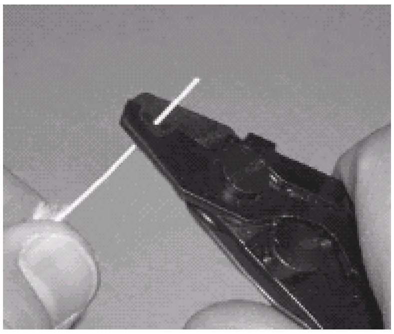
- Cut one meter of jacketed glass fiber optic cable. The fiber consists of core & clad of glass fiber, buffer coating, fibrous (Kevlar) material and outer jacket.



- Place the conical strain relief sleeve over the end of fiber such that broader end of sleeve should be kept towards the fiber tip.
- Remove the outer jacket coating of about 1 inch using fiber optic jacket stripper.
- Cut the fibrous (Kevlar) material using blade or scissors.



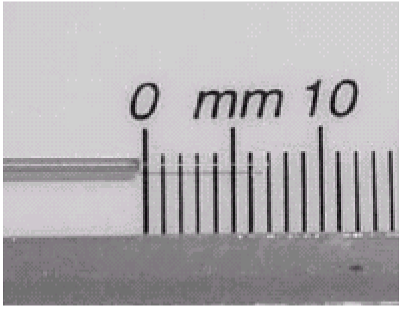
- Remove ½ inch of buffer coating using the fiber optic buffer stripper. Select the stripper hole of smaller diameter while removing the buffer. The buffer coating can also be removed chemically. Dip the fiber end in Methylene Chloride so that the coating get dissolved.



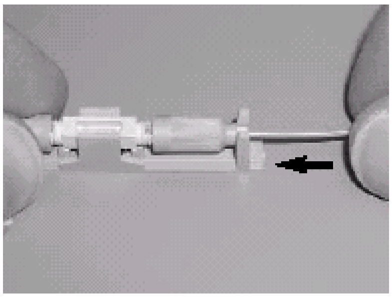
CAUTION: Care should be taken to avoid cutting the bare fiber of while trying to remove the outer and buffer coating.

- Wipe the bare fiber end with optic prep to remove the dust particles, which may have adhered to the fiber. To minimize losses one should ensure to remove all traces of impurities.

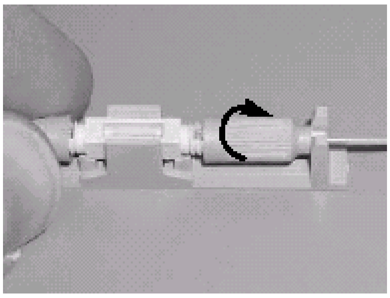
- Using a suitable cleaving tool to create a flat end face, cleave the expose fiber at approximately 7mm from the end of the buffer.



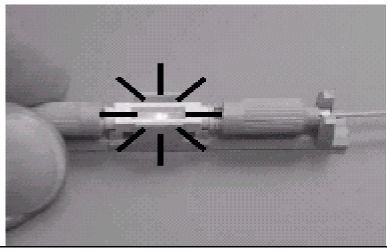
- Secure fiber by twisting the gray locking nut in the clockwise direction until it is finger tight. Be careful not to over-tighten, which can cause unnecessary attenuation. Repeat this procedure for the second fiber, which should contact the first fiber at the center of the glass capillary.



- Carefully insert fiber into one end of the ULTRA splice until it is centered in the view window. You should be able to visually inspect the fiber location in the glass capillary.



- It is recommended that a visual fault locator (VFL) be connected to one end of the fiber link for visual assurance of a good splice. The



light from the VFL will be visible through the ULTRA splice window and will fully extinguish upon completion of a good splice.

- If required, the ULTRA splice can be readjusted or “tuned” for optimal performance by untwisting the gray locking nut (counterclockwise) to make adjustments. Untwist one end only not more than a 1/2 turn. Gently pull back (slightly) and twist the buffer then push back against other fiber.

EXPERIMENT NO.4

EXPERIMENT NO-4

NAME

Measurement of Splicing Loss

OBJECTIVE

To Study Measurement of Splicing Loss1

EQUIPMENT

- 1 meter Glass Patch chord
- Optical Power Source
- Optical Power Meter

PROCEDURE

- Take one meter glass patch chord
- Connect one end of the patch chord to optical power source and other end to optical power meter.
- Switch on the power source and power meter. Select the wavelength in power meter in accordance with the wavelength of the source.
- Note the reading in power meter.
- Cut the fiber patch chord in two parts.
- Join these two parts using Ultra splice.
- Connect spliced patch chord to optical power source and optical power meter.
- Note the reading in power meter and find out the difference of the readings.
- The difference will denote Splice loss.